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Cybersecurity knowledge, social networking, and awareness among Gen Z university students

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Abstract

The Generation Z students have also become one of the most digitally active populations, and cybersecurity awareness is becoming very important to them. The aim of the current study is understanding the factors of cybersecurity awareness in Bangladesh and the mediating influence of risk perception. Data was collected among 398 university students in Dhaka, considering a stratified random sampling of both the public and the privately-owned institutions of higher learning, and analyzed using Partial Least Squares Structural Equation Modeling (SmartPLS 3.3.9). The findings suggest that cybersecurity knowledge is a strong factor that contributes to forming risk perception and awareness; social networking is also a contributing factor in the mentioned outcomes. Moreover, it is found that the relationship between cybersecurity knowledge and awareness works through risk perception but does not work through that of social networking and awareness. The results show that knowledge, social interaction, and personal perception together shape safe online behavior. This study gives practical advice for teachers and policymakers to improve youth cybersecurity. It adds new insights by focusing on Gen Z in a developing country during the post-pandemic period.

Keywords Cybersecurity knowledge, Social networking, Risk perception, Cybersecurity awareness, SEM, Bangladesh

1 Introduction

The contemporary speed of digital technologies development has reshaped much in the terms of communication in general, learning and business. With this transformation, automatization has opened an enormous opportunity, but it has also brought cyber threats, which endanger individuals, organizations, and society in general [4, 10, 43] (Lee and Kim 2023). Cybercrime has increased exponentially in the past ten years, and it is expected that in 2025, these losses can reach up to 10.5 trillion a year [23, 59]. The number of ransomware attacks is increased by more than 150% over the last five years, which indicates the complexity of malicious forces and the vulnerability of online systems



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(Sabillon et al. 2021) [52, 80]. In addition to technological infrastructures, human factors, unawareness, and unsafe user behavior can be considered the most significant dangers to cybersecurity breaches across the globe [2, 7, 47, 78].

The importance of cybersecurity has further increased in this age of Education 4.0, when digital platforms have become the major agent of teaching, learning, and collaboration. The COVID-19 pandemic contributed to the blistering pace even more, and online learning and remote work became essential in many areas that have internet access [5, 7, 28, 60, 65]. As online learning and work conditions become the industry norm, exposure to cyberattacks is increasing more and more, both on a personal and organizational level. The implications of breaches not only encompass monetary and data loss but a great deal of damage to academic reputation, institutional confidence, and student trust. Therefore, the promotion of cybersecurity consciousness has become one of the priorities around the world to maintain safe and efficient digital spaces [18, 21, 60, 62].

Generation Z has become an especially significant group to study in terms of cybersecurity within such a context. Grouped with those who were born between the late 1990s and early 2010s, this generation was brought up amidst digital fluency, multitasking skills and an extensive dependence on social media platforms of communication, research, and distraction [60, 66]. They are also more informed about wider social and environmental concerns and they tend to use social and visual forms of communication online (Lee & Kim 2023). This constant digital interaction, however, also increases their liability to cyber threats that include theft of their information to their abuse in the shape of cyber abuse and identity crimes [15, 28]. Cybersecurity knowledge and online behaviours are not only significant to academic success, but also to personal identity and social development within the digital domain especially to those who are university students [9].

In Bangladesh those concerns are especially acute. The nation has achieved tremendous success in terms of the scaled-up internet and digital transformation but at the same time, the dangers of the increased internet access are immense. In the first generation Z of university students, there is intensive use of several platforms for social networking, learning, education, and communication but there is lack of proper knowledge in regard to online security and risks. It is this loophole that poses risks to not just individuals but also bigger institutional and national systems to cyber vulnerabilities [20, 85]. In a bid to manage such risks the Bangladesh government has initiated essential measures that include the National Cybersecurity Strategy and the enactment of the Cybersecurity Operations Center (CSOC). Along with that, there is public awareness to encourage citizens towards digital responsibility [3, 85]. However, the development of cybercrimes in Bangladesh remains high, which confirms the necessity to enhance cybersecurity awareness, especially in youth, who are responsible to the largest extent of the digitalization of this country.

Although significant efforts are being made internationally and nationally to raise awareness around cybersecurity, there is limited research carried out on the topic especially in Bangladesh. The role of students has not been affirmed in the previous studies, yet higher education institutions emerge as the popular targets of data breaches and other types of assaults (Sabillon et al. 2021) [24, 57]. Additionally, little has been done in understanding the interactive relationship between individual factors-cybersecurity

knowledge, and social networking behaviors, and how such relationships inform cybersecurity awareness among people in their early adulthood years. The available literature also does not offer much information about the role of risk perception, even though the topic is central in driving protective behaviours online [27, 38]. These research areas leave gaps that require empirical studies that will aid in gathering knowledge of factors influencing responsible online behavior among Bangladesh university students.

To fill this gap, it is convenient to consider the Protection Motivation Theory (PMT). Originally proposed by Rogers (1975), PMT describes the reaction to the perceived perceived threat, as well as the factors that lead to protective behavior. The theory focuses on two processes, namely, threat appraisal, which is perception of danger, and coping appraisal, the process of measuring the capacity to effectively react to it. The use of PMT to study the Generation Z may result in a better understanding of the factors (perception of risk, knowledge, and capacity to social network) that modulate awareness of cybersecurity risks in this population group (Van Bavel et al. 2021) [27, 55]. This conceptualisation navigator can therefore form a solid guiding pillar to explore individual and contextual influences on online safety practices.

Based on this reasoning, the present study seeks to answer two research questions:

RQ1. What role do cybersecurity knowledge and social networking play in cybersecurity awareness?

RQ2. Does risk perception mediate the relationship between cybersecurity knowledge and social networking with cybersecurity awareness?

In answering these questions, the study would contribute in a number of ways. First, it helps fill a research gap to study the topic of cybersecurity awareness among the students of Generation Z in a developing-country context where digitalization is growing at an unprecedented pace but cybersecurity awareness is still an unexplored field. Second, it combines PMT and multivariate analysis to bring out the relationships among knowledge risk perception and recognition. Lastly, it is anticipated that the findings will inform policy action, educators, and institutions in Bangladesh across promoting cybersecurity awareness and educating young adults on how to have responsible conduct online.

2 Literature review, theoretical foundations and hypotheses development

2.1 Theoretical foundations

The study bases its premises on Protection Motivation Theory (PMT), Social Learning Theory, and cognitive-behavioral approaches to understand the behavior of Gen Z students as far as cybersecurity is concerned. In PMT, protective responses are expected to occur after a person cognitively evaluates the threats and becomes motivated to take action, this sentiment is also evidenced by our work on why cybersecurity knowledge is directly and indirectly related to increased awareness of risk (Anderson et al. 2019) [6]. The Social Learning Theory emphasizes the importance of observation and peer attitude, which suggests the idea that social networking can increase the risk perception by sharing the fatalities and discussing them. Cognitive-behavioral however elaborates on how the perceptions can be considered as psychological bridges between information and cognition [60].

Whereas the validity of PMT and Social Learning Theory has been increasingly proven in the developed markets where formal cybersecurity training and infrastructure have existed, the current study applied it in an emerging digital economy, namely, Bangladesh,

where the formal cybersecurity training is quite in substantial (Lee & Kim, 2023) [61]. The examples provided here, informal peer-to-peer interactions on social media sites such as Facebook or WhatsApp have a bigger role to play in generating awareness than the formal intervention in international research. By placing the theories in a regionalized perspective, the study emphasizes that the fundamental premises of the PMT and the Social Learning Theory would still be valid but they might orient differently in areas where there are emerging economies [17, 35]. This comparative lens simultaneously enhances the theoretical aptitude of the prospective model and accents the significance of situating behavior examination paradigms to various digital topography.

2.2 Cybercrime and the human aspects of cybersecurity

Cybercrime is on the rise and is not only expanding in volume but also in the weight of the crimes it is causing at an alarming rate; to the point that they have been predicted to surpass traditional crime in the nearest future (Bedser 2007; Anderson et al., 2019) [2]. Global estimates are staggering, reaching a cost of 600 trillion dollars in 2020—which is twice as high as reported in 2015 [58]. The COVID-19 pandemic put this trend on steroids and phishing alone increased by 600% in March 2020 (Lee & Kim 2023) [9]. These numbers reflect the extent of cyber risks that is unprecedented with no national boundaries.

Even with the improvements in the technological fronts on security, human factors have been found to continue being the weakest fronts of security. Cyberattacks in residential settings are close to 95% (Talib et al. 2010) [52] and many users remain untrained not only within the responsibilities of security but also the fundamentals of security [12]. Incompetence, negligence, and risky online behavior are often cited as the causes of breaches and serve as an indication of the importance of individuals as backdoors into the systems [16, 35, 61]. As a result, global cybersecurity breaches cost more than six hundred billion dollars in 2018, an indicator of the colossal expenses of human-based security weak points (Lee & Kim, 2023) [61].

These trends are fairly understood in the developed contexts, but are very different in the emerging digital economies like Bangladesh. In places like Australia, where mature cybersecurity support systems and training programs are not yet in place, even more crucial to the last line of defense is end-user themselves, especially Gen Z students [60]. Bangladesh does not have institutionalized education and training on various awareness programs related to the palliative care approach. The absence of institutionalized setting and training on palliative care training leads to an informal peer to peer learning through experience and social media [85]. This demonstrates both the weakness and strength of human factors: internationally understood as the weakest link, but at a local level playing the role of an adaptive channel of learning in low-resource settings [59]. To ensure that human factors of cybersecurity are taken care of, the technological means alone are not enough but it rather needs more context-specific approaches to developing awareness and resilience of young digital natives.

2.3 Cybersecurity in global and emerging economy contexts

The cybersecurity landscape of a nation relates not only to the robustness of its ICT infrastructure but also the socially and behaviorally inclined processes of people utilizing the system [49, 57]. Considering the fast development of digital technologies, cyber

threats are currently regarded as a considerable threat not only to economic growth in emerging economies but to the stability of the global financial system [60, 69]. Although developing countries have access to the digital sphere due to the increased internet penetration that surpassed the 10-percent mark, they have also become vulnerable in a new way [43, 49]. In poorer states, the limited awareness and inadequate infrastructures are exploited on an increasingly frequent basis by cybercriminals, who use weak cybersecurity practices to launch attacks that have cross-border implications [40]. That underlines the interconnectedness of global security when the vulnerabilities in one region can gradually lead to an increase in risks globally and, thus, cybersecurity is a global issue that cannot be ignored by governments, regional, and international bodies [40, 49, 60].

In such a case, the global-local interplay is especially important to countries like Bangladesh. On the one hand, the country experiences a high rate of ICT development, supported with the digital revolution and enhanced online attention of Gen Z learners [60]. Conversely, cybersecurity is not formally taught, or there are lesser institutional defenses, so the users are highly exposed to cyber-attacks [43]. As opposed to the developed economies where awareness programs are at an advanced stage, in Bangladesh, informal methods are used to disseminate cyber-related knowledge i.e. through peer-to-peer discussions and social networking. Therefore, whereas global literature tends to discuss cybersecurity in terms of infrastructures and rules, in the context of Bangladesh, the human and behavioral aspect will be the most important one. This makes cybersecurity expertise (understanding, risk assessment, awareness) at the user-level important to national resilience and to national resilience in broader global terms [17].

2.4 Cybersecurity knowledge, awareness, and risk perception

Cybersecurity knowledge is defined as the level of knowledge a person and their familiarity with their cybersecurity threats, safe practices, and technical expertise are necessary. The knowledge is based on the need to learn more and gain progressive knowledge in the area [61, 76]. On the other hand, cybersecurity awareness is described as the recognition of risks and the desire to execute preventative measures in dealing with such matters. Coming closely is a risk perception, i.e., the mental assessment of the probability of cyber threats and the consequences of their repercussions [2, 48, 76]. These constructs are interrelated numbers, as knowledge acts as the basis of the construct, awareness is the behavioral predisposition of preparations, and risk perception is the psychological suspense linking cognition to action. Having greater cost knowledge in providing cybersecurity theoretically increases awareness and risk perspectives. Knowledge teaches students how to recognize weaknesses and, therefore, makes them more sensitive to risks and more protective behaviors [27, 53, 72]. This line can be confirmed by global evidence: in technologically developed, such as the U.S. and Europe, knowledge is the consistent predictor of increased awareness [53, 86], yet the researchers advise that even basic literacy in dealing with digital tools does not guarantee sufficient protective behaviors [25, 43].

There are different empirical findings in different regions. Students in Saudi Arabia demonstrated low levels of awareness and a weak understanding of data protection [10, 27, 30], and in India, they had better insights into possible threats like phishing but still lacked, as the initial results showed, higher order protection skills [31, 34, 43, 43, 56, 73, 83]. Comparative researches conducted in both Turkey, Israel, Poland, and Slovenia

revealed that students were very knowledgeable, but they usually lacked the ability to implement the factors in the field [86]. On the same token, Nigerian students possessed fundamental knowledge though they could not defend their digital resources [33].

In the developing countries like Bangladesh, these fetishes are of more importance. Knowledge tends to be informal, learned through peers or social media or other means such as through lived experience, and hence has very limited impact to shape awareness and protective conduct. This points to the need to study the process of translating cybersecurity expertise to awareness and risk awareness in lowest-income environments where services are still moving towards institutionalization. Thus it was hypothesized that:

H1 Cybersecurity knowledge has a positive and significant effect on the cybersecurity awareness among GEN Z students.

H2 Cybersecurity knowledge has a positive and significant effect on the risk perception among GEN Z students.

2.5 Social networking, awareness, and risk perception

Social networking has redefined the online world of this generation Z and has influenced how people communicate, learn and interpret risks online [83]. In this research, it is presupposed that social networking will have a positive impact of great significance on cybersecurity awareness and risk perception [84, 86]. Compared to the past generations, generation Z was born in the age of digital technology and thus, has a seamless access to Facebook, Instagram, Twitter, and Snapchat as well as integrating these applications into their academic, social and personal life [7, 78, 86]. Although they allow collaboration, knowledge sharing, and even educational applications, the platforms also bring more risks to students in the form of oversharing, phishing, and identity theft [49, 56, 84].

In this study, the frequency of logging in or duration of using social networks is not conceptualized as social networking but the character of interaction since social networking is viewed as how students share, interact and respond to other participants in these digital communities [82]. Engagement with social media is generally dangerous to cybersecurity when they are carried out actively (e.g., disclosure of personal information, commenting, or sharing) compared to passively (e.g. browsing or observation). The literature notes that the Gen Z may want connectivity all the time, their peers will validate them, and they want to express themselves through social media; such the urge to achieve those things normalized potentially dangerous behavior, making students more exposed to risks despite being aware of the dangers of dangerous practices [33, 49, 81, 86]. Meanwhile, peer norms and pressure on decisions are also very strong, and in many cases stronger than formal charges by an institution in terms of educating individuals [86]. This demonstrates that social networking influences behavior through social and culture-based dynamics whereas cybersecurity knowledge lies in more of the technical and cognitive side of things.

Educational settings provide more evidence of these dynamics too. The studies have shown that educators often use social networking platforms in their classrooms by introducing assignments that imply the use of social media by students to read, discuss, or collaborate [7, 44, 84]. Social networking can therefore provide an avenue of interaction

between students and educators fostering collaborational-type interactions which can benefit to increase awareness towards cybersecurity practices [50, 67]. Nonetheless, the recreational purpose of the same platforms may expose students to personal information at a lower safety level concerning academic purposes [1, 33, 65, 84]. This combination of roles emphasizes the need to differentiate between why and how students use social networks, such as academic, professional, or recreational, since each of them implies variation in risk exposure [11, 84].

Therefore, social networking contrasts the knowledge of cybersecurity in the sense that it affects risk perception. Whereas cybersecurity knowledge provides students with knowledge about threats and protective measures, social networking places them in peer-based environments that influence whether or not those threats and protective measures are implemented [83, 84]. Therefore, the study will focus not only on the extent of social network usage but also on the characteristics and details of the engagement as the factors associated with a significant impact on the development of cybersecurity awareness and perception of risk among the Gen Z students. Thus it was hypothesized that:

H3 Social networking has a positive and significant effect on the cybersecurity awareness among GEN Z students.

H4 Social networking has a positive and significant effect on the risk perception among GEN Z students.

2.6 Risk perception as a mediating mechanism

Risk perception is widely recognized as a mediating psychological mechanism that links individuals' knowledge and social experiences to their awareness and protective behaviors online [51, 74, 76]. In global contexts, particularly in Western economies, empirical studies show that knowledge of cybersecurity threats alone does not necessarily translate into safe practices. Instead, it is the perceived severity and susceptibility of cyber risks that motivate precautionary action [48, 74, 80]. For instance, IoT users in Europe and the United States report stronger awareness only when their risk appraisal is high, aligning with Protection Motivation Theory's emphasis on threat appraisal as a driver of behavior [41].

In contrast, the regional context of Bangladesh and other emerging economies presents a different pattern. Gen Z students are digitally active and heavily engaged in social networking, but their knowledge is often acquired informally through peers rather than structured training programs [41, 74]. As a result, while baseline knowledge exists, their risk perception remains low, leading to unsafe behaviors such as oversharing, weak password use, and poor data protection. Prior studies in Saudi Arabia, Nigeria, and India similarly noted that knowledge without heightened perception of risk often fails to translate into meaningful cybersecurity awareness [10, 14, 33, 83] (Fig. 1).

This contrast between global (structured awareness campaigns) and local (peer-driven informal learning) contexts highlights the originality of the present study [41]. It emphasizes that risk perception is not merely an additional factor but a critical mediator in transforming knowledge and social networking into actionable cybersecurity awareness

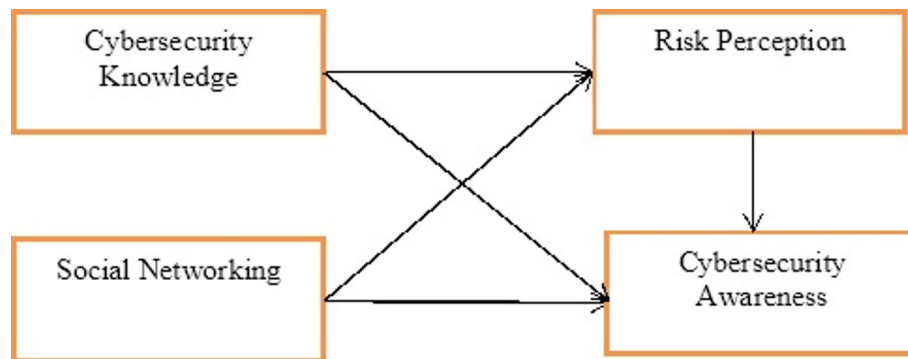


Fig. 1 The framework of the study

among Gen Z students in Bangladesh. Accordingly, this study hypothesizes that: Thus, it was hypothesized that:

H5 Risk perception has a positive and significant effect on the cybersecurity awareness among GEN Z students.

H6 Risk perception mediates the relationship between cybersecurity knowledge and cybersecurity awareness among GEN Z students.

H7 Risk perception mediates the relationship between social networking and cybersecurity awareness among GEN Z students.

3 Methodology of the study

3.1 Study context

In the present study, researchers adopted a positivist research strategy, a method that facilitates the acquisition of empirical knowledge by formulating hypotheses rooted in observations [63]. Furthermore, the research team performed an in-depth quantitative analysis as the concerns were highly specified, the assumptions were plainly stated, and all relevant information was considered [39]. Given the proactive stance of the Bangladeshi government in implementing a range of sustainability policies and tailor-made initiatives aimed at enhancing cybersecurity awareness among university students belonging to Generation Z, the selection of a sample from Bangladesh offers an enticing prospect for the execution of this investigation [64].

3.2 Sampling and data collection

Four hundred fifty questionnaires were distributed to students through institutional distribution from public and private universities in the Dhaka Division of Bangladesh who were a part of Generation Z, to obtain data for analysis. Online survey was also executed to make the data collection process smoothly. This data collection initiative was conducted over the span of September to October in the year 2024. The questionnaires used in this study were modified to better align with the specific context of Generation Z university students in Bangladesh. Adjustments included rephrasing questions for clarity, incorporating culturally relevant examples, and ensuring that the language resonated with the target demographic. These modifications aimed to enhance the relevance and effectiveness of the survey in capturing accurate data on cybersecurity awareness.

In line with the methodology of this cross-sectional study, a stratified random sampling approach was judiciously employed. Stratified random sampling was employed based on the type of university (public vs. private) to ensure representation across different educational institutions in Dhaka. This approach allows for a more comprehensive understanding of cybersecurity awareness among diverse student populations. Effective sample size refers to the number of respondents, whose data were usable for analysis after excluding incomplete or invalid responses, while response rate indicates the percentage of the total sample that completed the survey, reflecting the overall participation level. These distinctions are crucial for interpreting the study's findings accurately.

While the available pool of responses for potential inclusion in the sample numbered 398, 52 responses were subsequently excluded due to instances of ambiguity or missing information. Such omissions could potentially have been triggered by respondents' reservations about the survey's credibility or their preference for engaging solely in face-to-face interactions. Collectively, our findings meticulously align with the stipulated sample size criteria as set forth by Hair et al. [36]. The survey we conducted elicited a commendable response rate of 88.44% among the survey's distributed sample. Subsequently, the study's conclusions were derived from the scrutiny of data gathered from a cohort of 398 participants. Participants were comprehensively apprised of the confidential treatment bestowed upon their data, thereby ensuring that they willingly and voluntarily consented to their engagement in the study. Additionally, it was emphatically communicated that the collated data would be exclusively utilized for research purposes. As a precursor to the definitive survey administration, a pilot study involving a subset of 20 participants was executed; it is essential to underscore that these pilot participants were omitted from the final analysis of the survey results.

The research assistants undertook the task of collecting the data. The questionnaire was first given to five academicians and academic professionals in English to ensure that the items were clear and understandable. Every question in the questionnaire was scored from 1 to 5, meaning a score of 1 meant "strongly disagree" and 5 meant "strongly agree." We further checked whether the patterns of non-response among late respondents were similar to both the first 25% and the final 25% of late respondents. The results from the two tests show that we were not affected by non-response bias, consistent with Armstrong and Overton's [13] findings.

The demographic breakdown of the respondents is as follows: 65.63% identified as male, 37.37% fell within the age range of 22 to 25, 92.73% were single, 38.67% were engaged in the field of computer science, 22.45% were in their fourth year of college, 35.77% reported daily smartphone usage, and 56.33% possessed advanced-level computer skills.

3.3 Questionnaire design

There were four variables applied in the final survey, which contained 16 questions (excluding questions about demographics). The survey with 16 items meets the minimum standards set by Hair et al. [37]. To measure the cybersecurity knowledge construct, four items from Kennison and Chan-Tin [53] and Chandarman and Van Niekerk [18] were used. Then, Alharbi and Tassaddiq [7]'s four items were used to quantify social networking. In addition, four questions from Sarathchandra et al. [80] were used to test

the risk perception construct. To measure cybersecurity knowledge, Taha & Dahabiyeh's [73] four items were used.

Participants were asked to rate the constructs using Likert scale questions ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Cybersecurity awareness, risk perception, and social networking were measured using scales adapted from prior studies (see appendix). The total scores were computed by taking the average of each construct. PLS-SEM can use raw data, so it does not need to be standardized during the analysis. To ensure the reliability and validity of reflective constructs, we used Cronbach's alpha, composite reliability, and average variance extracted (AVE).

3.4 Data analysis

To validate the proposed research model, we employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS version 3.3.9, complemented by SPSS v25 for descriptive statistics and demographic profiling [36, 64]. PLS-SEM was chosen due to its suitability for exploratory research, predictive modeling, and its ability to handle complex structural models involving multiple constructs and indicators without requiring multivariate normality [37]. The model estimation involved analyzing both direct and indirect (mediated) effects through a series of ordinary least squares (OLS) regressions. To assess the significance of path coefficients and mediation effects, we applied a bootstrapping technique with 5000 resamples, a non-parametric method suitable for testing complex relationships.

A reflective measurement model was used to assess latent constructs, with evaluations conducted for indicator reliability, internal consistency, and construct validity. Specifically, Cronbach's alpha, Composite Reliability (CR), and Average Variance Extracted (AVE) were used to assess reliability and convergent validity, while discriminant validity was checked through standard criteria. We also examined convergence coefficients, variance inflation factors (VIF) to assess multicollinearity, and model fit through Standardized Root Mean Square Residual (SRMR) and R^2 values.

In addition to the SEM analysis, we applied a linear regression framework within PLS-SEM to investigate both direct relationships (e.g., from cybersecurity knowledge to awareness) and mediating effects (e.g., through risk perception). Although PLS-SEM does not inherently resolve reverse causality, this limitation was addressed through robust theoretical justifications, control variables (e.g., gender, age, year of study, daily used device, computer skills), and robustness checks. Future studies could further validate causal claims by employing advanced endogeneity tests, such as the Gaussian Copula approach. The minimum R-square criterion and sample-to-item ratio (5:1) were used to justify the adequacy of the sample size, with 398 respondents exceeding the minimum threshold [36]. Given PLS-SEM's proficiency in handling moderate sample sizes and simultaneously assessing measurement and structural models, it was deemed the most appropriate analytical technique for this study [36, 64].

4 Findings

4.1 Respondents' profile

Table 1 details the demographic characteristics of the respondents. The study rested on findings gained from gathering data from a total of 398 students who were part of Generation Z and were involved with various public and private universities in Dhaka, the

Table 1 Respondent's demographic profile

Characteristics	Frequency	Percentage	Characteristics	Frequency	Percentage
<i>Gender</i>			<i>Daily used device</i>		
Male	244	61.31	Smart phone	165	41.46
Female	154	38.69	Tablet	37	9.30
<i>Age</i>			Desktop	125	31.41
17–19 Years	177	44.47	Laptop	62	15.58
20–22 Years	111	27.89	Others	9	2.26
23–24 Years	43	10.80	<i>Computer skill</i>		
25–27 Years	27	9.73	Beginner	232	58.29
<i>Year of study</i>			Intermediate	69	17.34
1st Year	137	34.42	Advance	79	19.85
2nd Year	98	24.62	Others	18	4.52
3rd Year	122	30.65			
4th Year	29	7.29			
Internship year	12	3			
Total-398					

Table 2 Measurement of model assessment

Constructs	Items	Loading	AVE	CR	Alpha	R Square	NFI	SRMR
Cybersecurity knowledge	CK1	0.825						
	CK2	0.799	0.652	0.882	0.822			
	CK3	0.800						
	CK4	0.804						
Social networking	SN1	0.820						
	SN2	0.802	0.633	0.873	0.807			
	SN3	0.745						
	SN4	0.815						
Risk perception	RP1	0.811						
	RP2	0.852	0.623	0.868	0.798	0.404		
	RP3	0.778						
	RP4	0.711						
Cybersecurity awareness	CA1	0.766						
	CA2	0.807	0.629	0.871	0.802	0.638	0.732	0.901
	CA3	0.859						
	CA4	0.735						

AVE Average Variance Extracted, CR Composite reliability, NFI Normed Fit Index, SRMR Standardized Root Mean Square Residual

capital city of Bangladesh. Smart PLS 3.3.9 and SPSS V.25 were used as analysis tools during the research. This investigation embraced a pioneering approach to data collection. According to the data exhibited in Table 1, the respondents' profile indicates that 61.31% of the participants identified as male, 44.47% fell within the age bracket of 16–19 years, 34.42% were in the inaugural year of their academic journey, with a notable majority (41.46%) habitually utilizing smartphones on a daily basis. Furthermore, a significant proportion (58.29%) of respondents identified themselves as beginners in terms of computer proficiency.

4.2 Measurement, validity and reliability

Table 2 presents an evaluation of model performance. As per the heuristic guideline proposed by Vinzi et al. [77], outer loadings are recommended to be at least 0.50 or higher. In the context of this study, the external loadings, as indicated in Table 2, surpass the

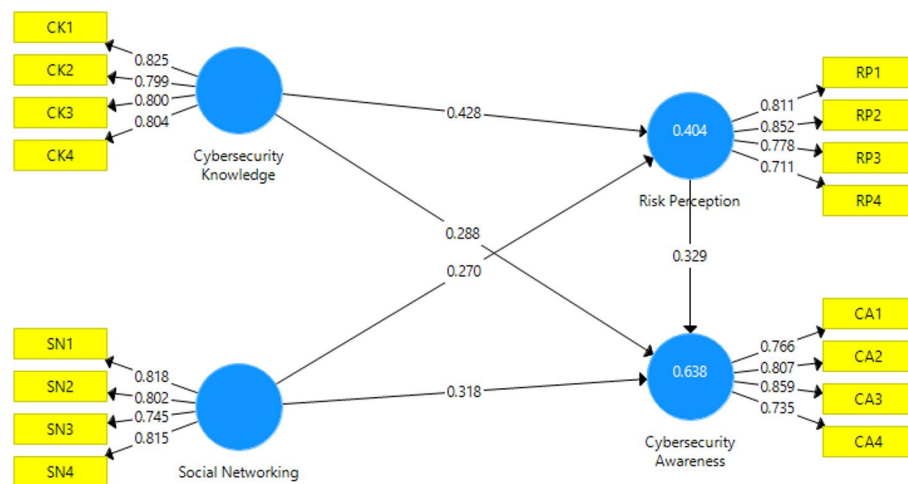


Fig. 2 Standardized results of SEM calculations

Table 3 Values of the stone geisser indicator (Q^2) and cohen's indicator (f^2) of the model in the SEM

Variables	Q^2	CA (f^2)	RP (f^2)
Cybersecurity awareness	0.383		
Cybersecurity knowledge	0.410	0.144	0.182
Risk perception	0.378	0.178	
Social networking	0.382	0.154	0.072

Large effect > 0.34; Medium effect > 0.14; Small effect > 0.01 (Cohen 1988)

CA Cybersecurity Awareness, RP Risk Perception

0.50 threshold. Factor loadings for the measurement items are visually represented in Fig. 2. Additionally, both the composite reliability (CR) and the scores from the Average Variance Extracted (AVE) test surpass the standard values of 0.7 and 0.5, respectively, denoting a dependable indication. Furthermore, in line with the guideline set by Hair et al. [37], Cronbach's alpha is found to exceed the minimal threshold of 0.70. The presence of a value higher than 0.70 for Cronbach's alpha for each construct, as elucidated in Table 2, underscores the empirical necessity for construct reliability [37]. Table 1 presents an exposition of the outcomes concerning construct reliability and the Average Variance Extracted (AVE). Table 2 delves into the potential influence of exogenous factors on cybersecurity awareness, accounting for 63.8% or 0.638. Correspondingly, these exogenous factors, exerting an influence of 40.4% (0.404), can wield a substantial impact on risk perception, as noted by Cohen and Squire [22]. The depiction in Table 2 also signifies that the Normed Fit Index (NFI) approaches unity, thus affirming the model's alignment with the stipulated research objectives. An SRMR (Standardized Root Mean Square Residual) score of 0.08 in the same table indicates a satisfactory fit of the model, a sentiment in consonance with the work of Hair et al. [37].

Table 3 outlines the Stone Geisser measure (Q^2) and Cohen's measure (f^2) for the structural equation model. Q^2 scores higher than zero indicate that the findings are still reliable and relevant for future uses [19]. Pertaining to effect sizes, it is apparent that cybersecurity awareness is subject to a moderate degree of influence from cybersecurity knowledge, social networking, and risk perception (f^2). In contrast, social networking and cybersecurity knowledge exhibit slight to moderate influences on risk perception,

Table 4 Fornell-Larcker criterion analysis for discriminant validity

		1	2	3	4
1	Cybersecurity awareness	0.793			
2	Cybersecurity knowledge	0.585	0.807		
3	Risk perception	0.456	0.598	0.790	
4	Social networking	0.434	0.542	0.489	0.795

The diagonal are the square root of the AVE (in bold) of the latent variables and indicates the highest in any column or row (Polas et al. 2022)

LV latent variable

Table 5 The heterotrait-monotrait (HTMT) analysis for discriminant validity

		1	2	3	4
1	Cybersecurity awareness				
2	Cybersecurity knowledge	0.776			
3	Risk perception	0.765	0.654		
4	Social networking	0.567	0.567	0.543	

Discriminant validity exists if the HTMT < 0.85 (Henseler et al. 2005)

respectively. In an overarching sense, the model demonstrates a satisfactory level of fitness and substantiates its robust predictive significance.

4.3 Discriminant validity

4.3.1 Fornell-Larcker criterion analysis

The Table 4 summarizes the assessment of discriminant validity using the Fornell-Larcker criterion. Table 4 presents the correlations among the inter-Latent Variables (LVs) and their corresponding of AVE values in the primary diagonal of the Structural Equation Model (SEM). Results from the Fornell-Larcker test [32] demonstrate reliable discriminant validity, concurring with the bolded square root of AVE. It should be noted that these coefficients are greater than the correlations between the variables, ranging from 0.790 to 0.807. By following this method, the model's ability to differentiate among variables is supported and accepted.

4.3.2 Heterotrait-Monotrait (HTMT) analysis

Table 5 presents the analysis of heterotrait-monotrait (HTMT) for assessing discriminant validity, as introduced by Chin [19]. In the process of cross-validating discriminant validity, it becomes evident that the HTMT values consistently remain below the threshold of 0.85, adhering to Chin's criteria [19]. Given that the HTMT value remains below this established threshold, the model's discriminant validity is statistically established, as outlined by Chin [19].

4.4 Structural model assessment

The validation process necessitates an assessment of the structural model's integrity. Additionally, through the utilization of the bootstrapping technique with 5000 resamples, T-values and R^2 values were computed for further analysis (Polas et al. 2022). The outcomes of the structural equation modeling (SEM) computations conducted using Smart PLS 3.3.9 is visually represented in Fig. 2. Across the board, the observed external loads of the variables frequently surpassed the acceptable thresholds. Moreover, a majority of the variables exhibited highly favorable directional coefficients (Polas et al. 2022).

Table 6 Results of direct and indirect effect hypotheses

Hypotheses	Relationship	Std Beta	Std error	t-value	p-value	Decision
H1	Cybersecurity Knowledge → Cybersecurity Awareness	0.288	0.123	2.338	0.020	Supported
H2	Cybersecurity Knowledge → Risk Perception	0.421	0.126	3.400	0.001	Supported
H3	Social Networking → Cybersecurity Awareness	0.318	0.105	3.042	0.002	Supported
H4	Social Networking → Risk Perception	0.286	0.115	2.349	0.019	Supported
H5	Risk Perception → Cybersecurity Awareness	0.329	0.151	2.172	0.030	Supported
H6	Cybersecurity Knowledge → Risk Perception → CA	0.139	0.079	1.987	0.048	Supported
H7	Social Networking → Risk Perception → CA	0.094	0.059	1.498	0.135	Rejected

CA cybersecurity awareness

4.5 Hypotheses testing (direct and indirect relationships)

As Table 5 shows, cybersecurity knowledge contributes marginally to increase awareness ($\beta = 0.288$, $t = 2.338$, $p < 0.05$, refer to Table 6) and thus, supports Hypothesis 1. The finding aligns with previous literature [18, 53], which underlines that educated people could better recognize the dangers and exercise safer online practices. Among the Gen Z group in Bangladesh, knowledge is especially important as it corresponds to the needs of the demographic as digital natives with little or no formal access to cybersecurity training. Additionally, internationally, studies demonstrate that formal knowledge in a form of seminars, workshops or simulations can convert into increased consciousness and safer behaviours both in learning institutions and organizations. Our findings therefore broaden existing theories of knowledge-behavior linkage to a third-world setting indicating the global significance of knowledge as a source of awareness. Practical interventions should be adopted to reinforce awareness via a direct transfer of knowledge.

As the results reveal, cybersecurity knowledge has a positive and significant influence on risk perception ($\beta = 0.421$, $t = 3.400$, $p < 0.05$, consult Table 6), which brings Hypothesis 2 supported. This is aligned with the previous works [12, 76], which reveal that more knowledgeable individuals feel more vulnerable to cyber threat. This finding provides evidence that knowledge indeed is a key factor in determining whether people are risk sensitive or not, especially in the Bangladesh experience of Gen Z where digital literacy and infrastructures are still at the initial phases. Contrary to what may apply to Western settings where there is already some awareness on cyber risks, in Bangladesh, students need institutional efforts to raise awareness among them. Conceptually, this paradigm emphasizes the mental circuit to which knowledge leads to the risk appraisal which is a precondition to safe practice adoption. In practice, it implies that there is a way to increase the feeling of vulnerability and foster protective actions through workshop, simulations, and other local interventions. Our findings not only expand the knowledge-risk perception nexus to a new realm—an emerging economy and a digital-native generation—but also help unpack the conceptualization of knowledge and its role in risk perception.

The results are in line with Hypothesis 4 because it is evident that social networking makes a significant difference in risk perception ($\beta = 0.286$, $t = 2.349$, $p < 0.05$, refer to Table 6). This corresponds to what previous studies have revealed [12, 76] that online communication exploits perceptions of cyber threats. In the case of Bangladesh Gen Z,

since formal cybersecurity is given less importance, informal peer communication is one of the sources of awareness, through Facebook, WhatsApp, or TikTok. Students are likely to learn about hazards through shared experiences or posts as well as a result of exposure to cyber incidents that have been discussed in online forums. On a worldwide basis, this has represented Social Learning Theory in which observation and interaction create awareness. Although social media can create misinformation, it also increases the vigilance level because it is now normal to discuss risks. In this way, the paper extrapolates international results to a developing world nation as the social networking plays the role of an education tool on the grass-root level of digital-native risk awareness.

The findings indicate that risk perception has a positive and significant effect on cybersecurity awareness ($\beta = 0.329$, $t = 2.172$, $p < 0.05$, consult Table 6), which means that Hypothesis 5 has been confirmed. This is consistent with previous research [48, 79] as well as the Protection Motivation Theory that postulates that threats to safety lead to defense motivation and action. In the case of young people in Bangladesh, Gen Z, part of whose country does not receive comprehensive training in cybersecurity, self-awareness is usually created when the student feels that they are threatened by a dangerous situation like phishing or identity theft. This has a global cognitive connection with perception and awareness: Being alert promotes learning and safer performance. The finding relates to the previous hypotheses, proving that although knowledge enhances the development of awareness and perception, perception is also a cognitive stimulator of awareness. Therefore, in both emerging digital economies and in developed places, the narrative of overall digital vulnerability can act as a boost to student awareness and knowledge.

Findings also reveal that the risk perception mediates the relationship between cybersecurity knowledge and awareness ($\beta = 0.139$, $t = 1.987$, $p < 0.05$, see Table 6), validating Hypothesis 6. This indicates that knowledge has a direct and indirect effect on increasing awareness through influencing the perceptions of the learners on the threats of online activities. In accordance with Protection Motivation Theory (PMT), awareness develops when the knowledge is comprehended as the perceived risk and drives people to act in a protective way [48, 79]. This pathway is particularly significant in Bangladesh Gen Z, since low levels of formal training translates to predominant self-reliant identification of risk. In contrast to the western literature whereby structured programs could mediate this gap, peer-sharing and experience learning is more instrumental in the development of risk perception in the South Asian and other developing environments. The finding combines previous hypotheses (H1, H2), proving that knowledge can change awareness directly and indirectly through the cognitive-affective path of perceived vulnerability, further supporting the necessity of risk-conceptualized digital education on the international scale.

Hypothesis 7 proposed that risk perception mediates the relationship between social networking and cybersecurity awareness. However, the mediation effect was not supported ($\beta = 0.094$, $t = 1.498$, $p > 0.05$, see Table 6). This indicates that while social networking directly enhances awareness (as supported in H3), it does not do so indirectly through risk perception.

Theoretically, the Hypothesis 7 can be rejected meaning that the daily use of social networking, specifically in Bangladesh, does not promote risk perception by Gen Z.

Peer-imposed atmospheres can turn hazardous behavior to be typical, as opposed to structured learning situations that raise hazard sensitivity [79]. This points to the edge in terms of digital interaction and effective security thoughtfulness, which, in turn, bring about the necessity to direct cybersecurity training and practical interventions.

5 Discussion

This study aimed to examine how cybersecurity knowledge and social networking influence cybersecurity awareness and risk perception among Generation Z university students in Bangladesh. It also investigated the mediating role of risk perception in these relationships. Addressing a notable research gap, the study focused on students in a developing country context, where online activity is growing rapidly but cybersecurity literacy remains underexplored. To explore these relationships, the study employed a quantitative method using Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze data collected from 398 undergraduate students enrolled in both public and private universities in Dhaka, Bangladesh.

The results revealed that cybersecurity knowledge significantly influences both risk perception and cybersecurity awareness, affirming that students who are more knowledgeable about online security risks tend to develop stronger awareness of cybersecurity threats. Similarly, social networking was found to have a positive and significant effect on both cybersecurity awareness and risk perception, though the strength of influence was slightly lower compared to cybersecurity knowledge. Importantly, risk perception was found to mediate the relationship between cybersecurity knowledge and cybersecurity awareness, but not the relationship between social networking and awareness indicating that the awareness gained through social media interactions might not always stem from a well-developed understanding of online risks.

These findings have important social policies. First, they suggest that enhancing students' knowledge of cybersecurity threats directly improves their risk perception and, subsequently, their awareness and behavioral readiness. Second, while social networking plays a key role in shaping communication and peer interaction, it alone may not sufficiently raise risk perception unless supported by structured educational initiatives. Therefore, incorporating cybersecurity awareness programs into academic curricula, student workshops, and online safety campaigns could be instrumental in bridging the gap between awareness and actual protective behavior. Moreover, the study uncovers the critical distinction between knowing about threats and acting to mitigate them. Many students, despite being aware of security risks, do not consistently apply security measures such as strong password practices or regular software updates. This highlights a pressing need for behavioral interventions that move beyond awareness to foster actionable cybersecurity habits.

The contribution of this study lies in offering a comprehensive empirical model that captures the interplay between knowledge, social influence, risk perception, and cybersecurity awareness in a developing country context. It provides both theoretical insights and practical strategies for educators, policymakers, and cybersecurity professionals aiming to improve students' digital safety. By prioritizing risk understanding and peer-influenced learning, institutions can develop targeted programs that better protect Generation Z students from evolving online threats. This framework may serve as a

foundation for future research and policy development aimed at fostering a more cyber-resilient academic environment.

6 Conclusion and implications of the study

This paper is one of the few empirical studies examining the influence of cybersecurity knowledge and social networking on risk perception and awareness among the Gen Z students in Bangladesh because digital adoption is on the rise but the literacy level is uneven. Based on the Protection Motivation Theory, and the Social Learning Theory, the study points out that the knowledge-driven cognitive processes and the socially-driven awareness have different pathways. The evidence supports the point that even though students have a high level of engagement on web, their safe practices tend to trail their knowledge hence creating a severe void between risk perception and safe practice. In this way, the study contributes to the global discussion of cybersecurity by placing these dynamics in the context of a developing country, providing not only theoretical complement but also empirical advice on ways to encourage cyber resilience in the youth going forward.

Theoretically speaking, the study makes a contribution in three ways. First, it updates the Protection Motivation Theory to a setting in a developing country and demonstrates that the knowledge-to-risk-perception channel will work robustly and perhaps even more effectively with low literacy levels. Second, it fills in the gap of boundary conditions in the Social Learning Theory showing how social influence raises both topic awareness and salience but not risk perception in real-life peer settings. Third, it suggests the presence of a dual-pathway that leads to a distinction between salience-driven monitoring leading to awareness of a social influence and appraisal-driven monitoring resulting in awareness of knowledge acquisition, thus explaining why most students end up knowing of a threat but not taking any action.

Practically, the results indicate that it is necessary to provide greater emphasis on cybersecurity knowledge in curricula and awareness causes because it is the most dependable motivator of risk perception in addition to awareness. Meanwhile, social networking cannot be ignored outright but needs to be considered through a strategic lenses by integrating reflective, risk-contextualized content on social media, university and policymakers can potentially translate salience into deeper appraisal. To bridge the knowledge-behavior gap that continues to exist between knowledge dissemination and the subsequent practice, interventions need to be beyond the informative scope and engage in reinforcing protective habits, i.e., two-factor authentication or secure password management, consolidation through actionable scripts as well as peer-led initiatives.

Hence, the analysis proves that cybersecurity knowledge and social networking have one-of-a-kind but integrated roles in developing cybersecurity awareness among the Bangladeshi students. Knowledge works through a cognitive channel and expands the risk perception and thus builds awareness, whereas social networking causes the awareness to increase but does not further the other perceptions and feelings of vulnerability. This dual pathway perspective offers valuable contributions to theory and practice, giving us an idea of how human factors may be informed in the developing contexts and how digital safety may be promoted among a generation that would play a crucial role in the economic and technological development in the future.

7 Limitations and future directions

A number of limitations to this research guide future research. One is the cross-sectional survey based on a sample of 398 students that study in Dhaka universities, which limits the causal meaning of the works and geographical generalization. Longitudinal or experimental studies, along with rural and regional institution sampling, would give a more comprehensive outline of how awareness and protective-behavior emerge in Bangladesh. Second, the research focused on cybersecurity knowledge and social networking as the key antecedents whereas other important antecedents were not considered: cultural norms, institutional support, and digital self-efficacy. The inclusion of such factors would make the model richer and give a more comprehensive picture of security behavior. Third, awareness and risky practices could be exaggerated due to social desirability bias since students might be more self-reporting than honest. The future research should adopt mixed-methodologies studies or behavioral statistics to allow confirmation of the findings. Last, the study targeted Generation Z in Bangladesh, thus, Africa and other generations may not generalize. Comparative studies across the Generations X versus Y and cross-national contexts would help determine whether the two pathways identified in this research, knowledge-based appraisal versus social influence-based salience, can be replicated in different groups. Overall, the complementary aspect of broadening the sample, deeper variables, and the multi-method-based approach should help reinforce both theoretical improvements and practical approaches to improving cybersecurity resilience.

Appendix

Measurement Scale

1. Cybersecurity knowledge [18, 53]
 - a. I am concerned/ have knowledge regarding Cyberbullying.
 - b. I am very careful during downloading, sharing and use of pirated content,
 - c. My knowledge of password security is high.
 - d. I know a lot about password security practices.
2. Social networking [7]
 - a. It is acceptable to post personal pictures on social media.
 - b. It is ok to accept friend requests from strangers.
 - c. There is no problem with sharing my current location publicly on social media.
 - d. I know how to report any threat or suspicious activity on social media.
3. Risk perception [80]
 - a. I am confident I can identify a phishing email or a social engineering attack.
 - b. I am adequately protected from malware, scareware and spyware.
 - c. I am well liked and will never be a victim of cyberbullying.
 - d. My passwords are strong enough. Nobody can guess them.
4. Cybersecurity awareness [73]
 - a. I'm aware of the Information Security concept.

- b. I'm aware of the Malware concept.
- c. I'm aware of the Antivirus software concept.
- d. I know that using https protocol is more secure than using http protocol in the URL.

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B. A. and M.R.H.P. wrote the main manuscript text, and M.F. prepared tables and figures. R.K. and M.I.T. drafted and revised the manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are not publicly available due to confidentiality of respondents that we assured them during data collection period but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Approval (Code-CRTC/2024/01) was obtained from the ethics committee of Sonargaon University (SU). The procedures used in this study adhere to the tenets of the Declaration of Helsinki. Informed consent was obtained from all individual participants included in the study.

Competing interests

The authors declare no competing interests.

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